

`nushell` structured terminal output for agent harness

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Summary

JSON for terminal output pairs with bash tool & agent skills enable error log analysis & feedback loop automation.

```
brew install nushell
```

Example: logs/app.log

Typical mixed-format application log

```
2025-05-03T10:15:22Z INFO   AuthService User login successful user_id=48392
2025-05-03T10:15:23Z WARN   CatalogService Missing attribute: color sku=SKU12345
2025-05-03T10:15:24Z ERROR  CatalogService Failed to enrich product sku=SKU98765 reason="timeout"
2025-05-03T10:15:25Z INFO   PriceService Price updated sku=SKU12345 old=199 new=179
2025-05-03T10:15:27Z ERROR  RAGService Vector search failed query="wireless earbuds" code=503
```

Nushell parsing command

```
open logs/app.log  
| lines  
| parse "{timestamp} {level} {service} {msg}"  
| where level = "ERROR"  
| to json
```

Example: Nushell JSON output (after filtering)

```
[
  {
    "timestamp": "2025-05-03T10:15:24Z",
    "level": "ERROR",
    "service": "CatalogService",
    "msg": "Failed to enrich product sku=SKU98765 reason=\"timeout\""
  },
  {
    "timestamp": "2025-05-03T10:15:27Z",
    "level": "ERROR",
    "service": "RAGService",
    "msg": "Vector search failed query=\"wireless earbuds\" code=503"
  }
]
```

Random thoughts starter

- Agent harness such as OpenClaw / Hermes Agents supports bash for terminal commands.
- LLM works well with structured outputs.
- Pairs with agent skills for log analysis & automations.